

A 24-CELL WEAK 4-RESOLVING SET OF $K_{11} \times K_{11}$ REFUTES A SURMISED EXACT VALUE

ABSTRACT. For a connected graph G and an integer $k \geq 1$, a set $S \subseteq V(G)$ is *weak k -resolving* if $\Delta_S(x, y) := \sum_{z \in S} |d(x, z) - d(y, z)| \geq k$ for every pair of distinct vertices x, y , and $\text{wdim}_k(G)$ is the least size of such a set. Farhan, Kuziak and Yero prove that $\text{wdim}_4(K_n \times K_n) \leq 2n + 1 + \lfloor n/4 \rfloor$ for every $n \geq 9$ and wonder, in the first item of their concluding remarks, whether this upper bound is the exact value. It is not. We exhibit a 24-cell weak 4-resolving set of $K_{11} \times K_{11}$, prove the matching lower bound $\text{wdim}_4(K_n \times K_n) \geq 2n + \lceil 2n/11 \rceil$ for every $n \geq 4$, and conclude

$\text{wdim}_4(K_n \times K_n) = 2n + \lfloor n/4 \rfloor = 24, 27, 29, 31$ for $n = 11, 12, 13, 14$,
one less than the surmised 25, 28, 30, 32 in each case, together with the
infinite family $\text{wdim}_4(K_{11m} \times K_{11m}) = 24m$ for every $m \geq 1$. The upper
bound of the source is true and is not contradicted.

1. INTRODUCTION

All graphs are finite, simple and connected. For vertices x, y of G and $S \subseteq V(G)$ put

$$\Delta_S(x, y) = \sum_{z \in S} |d(x, z) - d(y, z)|.$$

Following Peterin, Sedlar, Škrekovski and Yero [3], S is a *weak k -resolving set* of G if $\Delta_S(x, y) \geq k$ for every pair of distinct $x, y \in V(G)$, and the *weak k -metric dimension* $\text{wdim}_k(G)$ is the smallest cardinality of a weak k -resolving set. The condition is imposed on *every* pair, including pairs of vertices that lie in S .

The *direct* (tensor, categorical) product $G \times H$ has vertex set $V(G) \times V(H)$, with (g, h) adjacent to (g', h') if and only if $gg' \in E(G)$ and $hh' \in E(H)$. Throughout, $n \geq 3$ and we identify $V(K_n \times K_n)$ with the cells of an $n \times n$ array, $[n] \times [n]$, written (row, column) and indexed from 1. For $n \geq 3$ the graph $K_n \times K_n$ is connected of diameter 2, and for $u = (i, j)$, $v = (i', j')$,

$$(1) \quad d(u, v) = \begin{cases} 0, & u = v, \\ 1, & i \neq i' \text{ and } j \neq j', \\ 2, & \text{otherwise (exactly one coordinate agrees)}. \end{cases}$$

The Cartesian analogue is settled: $\text{wdim}_4(K_n \square K_n) = 2n$, a special case of the treatment of Hamming graphs in [2].

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The statement under test. A. Farhan, D. Kuziak and I. G. Yero, *The weak k -metric dimension of the direct product of complete graphs* [1], determine $\text{wdim}_k(K_n \times K_n)$ for every $n \geq 4$ and every admissible k except $k = 4$, where they prove an upper bound. In the notation of [1], Theorem 3.3 reads

$$(2) \quad \text{wdim}_4(K_n \times K_n) \leq 2n + 1 + \left\lfloor \frac{n}{4} \right\rfloor \quad \text{for every integer } n \geq 9,$$

and the first item of the concluding remarks of [1], introduced by “the following open questions might be of interest”, reads verbatim

“Since the case $k = 4$ is the only one in which an exact formula was not deduced, it is clearly of interest to complete this case, and we wonder on whether the exact value is precisely that shown as an upper bound in Theorem [3.3].”

The bracketed number replaces a cross-reference of the source, which resolves to Theorem 3.3, i.e. to (2). Resolved into a quantified statement, what is surmised is

$$(3) \quad \text{wdim}_4(K_n \times K_n) = 2n + 1 + \left\lfloor \frac{n}{4} \right\rfloor \quad \text{for every integer } n \geq 9.$$

It carries no statement number in [1], which is why it is quoted in full rather than cited by number.

Results.

Theorem 1.1. *The set of 24 cells*

$$(4) \quad S = \{(1, 1), (1, 10), (2, 1), (2, 11), (3, 2), (3, 10), (4, 2), (4, 11), \\ (5, 3), (5, 10), (6, 3), (6, 11), (7, 4), (7, 5), (8, 6), (8, 7), \\ (9, 8), (9, 9), (10, 4), (10, 6), (10, 8), (11, 5), (11, 7), (11, 9)\}$$

is a weak 4-resolving set of $K_{11} \times K_{11}$. Hence $\text{wdim}_4(K_{11} \times K_{11}) \leq 24 < 25 = 2 \cdot 11 + 1 + \lfloor 11/4 \rfloor$, and (3) is false.

Theorem 1.2. *For every integer $n \geq 4$,*

$$\text{wdim}_4(K_n \times K_n) \geq 2n + \left\lceil \frac{2n}{11} \right\rceil.$$

Theorem 1.3. *For $n = 11, 12, 13, 14$,*

$$\text{wdim}_4(K_n \times K_n) = 2n + \left\lfloor \frac{n}{4} \right\rfloor = 24, 27, 29, 31,$$

which is exactly one less than the value 25, 28, 30, 32 surmised in (3). Moreover $\text{wdim}_4(K_{11m} \times K_{11m}) = 24m$ for every integer $m \geq 1$, whereas (3) gives $22m + 1 + \lfloor 11m/4 \rfloor$ there; the gap $1 + \lfloor 11m/4 \rfloor - 2m$ grows linearly, like $3n/44$.

2. THE DISTANCE SUMS, AND A TWO-LINE CRITERION

Let $S \subseteq [n] \times [n]$. Write $a_i = |\{j : (i, j) \in S\}|$ for the i -th row degree and $b_j = |\{i : (i, j) \in S\}|$ for the j -th column degree, so $\sum_i a_i = \sum_j b_j = |S|$. For distinct $x = (i, j)$ and $y = (i', j')$ put $\alpha = |\{x, y\} \cap S|$, and when $i \neq i'$ and $j \neq j'$ also $\beta = |S \cap \{(i, j'), (i', j)\}|$ — the number of the two remaining corners of the rectangle spanned by x and y that lie in S .

Lemma 2.1. *Let $n \geq 3$ and $x = (i, j) \neq y = (i', j')$ in $K_n \times K_n$. Then*

$$\Delta_S(x, y) = \begin{cases} b_j + b_{j'} + \alpha, & i = i', \\ a_i + a_{i'} + \alpha, & j = j', \\ a_i + a_{i'} + b_j + b_{j'} - \alpha - 2\beta, & i \neq i' \text{ and } j \neq j'. \end{cases}$$

This is Proposition 2.2 of [1]; we include the proof to keep the note self-contained.

Proof. Throughout, $z = (p, q)$ ranges over S and distances are read off (1).

Suppose first $i = i'$ and $j \neq j'$. If $z = x$ then $d(x, z) = 0$ while z and y share their row, so $d(y, z) = 2$ and the term is 2; symmetrically $z = y$ contributes 2. For $z \notin \{x, y\}$: if $p = i$ then z shares its row with both x and y , so $d(x, z) = d(y, z) = 2$ and the term is 0; if $p \neq i$ then $d(x, z) = 2$ exactly when $q = j$ and $d(y, z) = 2$ exactly when $q = j'$, so the term is 1 precisely when $q \in \{j, j'\}$ and 0 otherwise. The cells of S with $q = j$ and $p \neq i$ number $b_j - [x \in S]$, and those with $q = j'$ and $p \neq i$ number $b_{j'} - [y \in S]$. Altogether $\Delta_S(x, y) = 2\alpha + (b_j - [x \in S]) + (b_{j'} - [y \in S]) = b_j + b_{j'} + \alpha$. Exchanging the roles of the two coordinates gives the case $j = j'$.

Now let $i \neq i'$ and $j \neq j'$, so $d(x, y) = 1$. If $z = x$ the term is $|0 - 1| = 1$, and likewise for $z = y$; these contribute α . For $z \notin \{x, y\}$ we have $d(x, z), d(y, z) \in \{1, 2\}$, and $d(x, z) = 2$ exactly when z shares a coordinate with x , i.e. when $p = i$ or $q = j$. Hence the term is 1 precisely when z shares a coordinate with exactly one of x, y . The cells of S sharing a coordinate with x are $a_i + b_j - [x \in S]$ in number, of which x itself is one when $x \in S$, so $a_i + b_j - 2[x \in S]$ of them lie in $S \setminus \{x, y\}$ (note y shares no coordinate with x); symmetrically for y . A cell sharing a coordinate with both x and y satisfies $(p = i \text{ or } q = j)$ and $(p = i' \text{ or } q = j')$; as p cannot equal both i and i' and q cannot equal both j and j' , it is (i, j') or (i', j) , so there are β of them and none is x or y . By inclusion–exclusion the number of $z \in S \setminus \{x, y\}$ sharing a coordinate with exactly one of x, y is

$$(a_i + b_j - 2[x \in S]) + (a_{i'} + b_{j'} - 2[y \in S]) - 2\beta = a_i + a_{i'} + b_j + b_{j'} - 2\alpha - 2\beta,$$

and adding the contribution α of $z \in \{x, y\}$ gives the third case. $\square$

Let G_S be the bipartite graph with parts $\{r_1, \dots, r_n\}$ and $\{c_1, \dots, c_n\}$ in which $r_i c_j$ is an edge if and only if $(i, j) \in S$; thus $|E(G_S)| = |S|$, $\deg r_i = a_i$ and $\deg c_j = b_j$. For a subgraph F of G_S write $T(F)$ for the sum of the G_S -degrees of the vertices of F . Rectangles with three corners in S are exactly

the paths with three edges in G_S , and rectangles with four corners in S are exactly the 4-cycles of G_S .

Lemma 2.2. *Let $n \geq 3$ and let $S \subseteq [n] \times [n]$ have $a_i \geq 2$ and $b_j \geq 2$ for all i, j . Then S is a weak 4-resolving set of $K_n \times K_n$ if and only if*

- (P) $T(P) \geq 9$ for every path P with three edges in G_S , and
- (C) $T(C) \geq 10$ for every 4-cycle C in G_S .

Proof. Pairs with $i = i'$ or $j = j'$ are settled at once: by Lemma 2.1 their Δ_S is at least $b_j + b_{j'} \geq 4$, resp. $a_i + a_{i'} \geq 4$, and they involve neither (P) nor (C). So only pairs with $i \neq i'$ and $j \neq j'$ matter, where Lemma 2.1 gives $\Delta_S(x, y) = T - \alpha - 2\beta$ with $T = a_i + a_{i'} + b_j + b_{j'} \geq 8$, and the requirement $\Delta_S(x, y) \geq 4$ reads

$$(5) \quad \alpha + 2\beta \leq T - 4.$$

Here $\alpha + \beta$ is the number of corners of the rectangle spanned by x, y that lie in S , and α counts those on the diagonal $\{x, y\}$. If $\alpha + \beta \leq 2$ then $\alpha + 2\beta \leq 4 \leq T - 4$ and (5) holds. If $\alpha + \beta = 3$ then either $(\alpha, \beta) = (2, 1)$, when $\alpha + 2\beta = 4 \leq T - 4$ holds automatically, or $(\alpha, \beta) = (1, 2)$, when (5) says exactly $T \geq 9$. If $\alpha + \beta = 4$ then $(\alpha, \beta) = (2, 2)$ and (5) says exactly $T \geq 10$.

Thus S is weak 4-resolving if and only if every rectangle with exactly three corners in S has $T \geq 9$ and every rectangle with four corners in S has $T \geq 10$. The second condition is (C). Given (C), the first condition is equivalent to (P): a three-edge path whose fourth rectangle corner also lies in S spans a 4-cycle, for which (C) already gives $T \geq 10 \geq 9$; and a three-edge path whose fourth corner is outside S is a rectangle with exactly three corners in S , with the same T . $\square$

3. PROOF OF THEOREM 1.1

The 24 cells of S in (4) are distinct, and the row and column degree sequences are

$$(a_1, \dots, a_{11}) = (b_1, \dots, b_{11}) = (2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3),$$

summing to $9 \cdot 2 + 2 \cdot 3 = 24$ both ways. In particular every degree is at least 2, so Lemma 2.2 applies. The row neighbourhoods in G_S are

$$(6) \quad \begin{aligned} N(r_1) &= \{c_1, c_{10}\}, & N(r_2) &= \{c_1, c_{11}\}, & N(r_3) &= \{c_2, c_{10}\}, \\ N(r_4) &= \{c_2, c_{11}\}, & N(r_5) &= \{c_3, c_{10}\}, & N(r_6) &= \{c_3, c_{11}\}, \\ N(r_7) &= \{c_4, c_5\}, & N(r_8) &= \{c_6, c_7\}, & N(r_9) &= \{c_8, c_9\}, \\ N(r_{10}) &= \{c_4, c_6, c_8\}, & N(r_{11}) &= \{c_5, c_7, c_9\}. \end{aligned}$$

Condition (C) holds vacuously: G_S has no 4-cycle. A 4-cycle is a pair of rows sharing two columns, so it suffices that any two of the eleven sets in (6) meet in at most one element. Split the columns into the block $B = \{c_1, c_2, c_3\}$, the block $B' = \{c_{10}, c_{11}\}$ and the block $M = \{c_4, \dots, c_9\}$.

Each of $N(r_1), \dots, N(r_6)$ consists of one column of B and one column of B' , and the six pairs so formed are the six distinct elements of $B \times B'$; two distinct pairs agree in at most one coordinate. Each of $N(r_7), \dots, N(r_{11})$ lies inside M , hence is disjoint from every $N(r_i)$ with $i \leq 6$. Finally $N(r_7), N(r_8), N(r_9)$ are pairwise disjoint, $N(r_{10})$ and $N(r_{11})$ are disjoint, and $N(r_{10})$ meets each of $N(r_7), N(r_8), N(r_9)$ in exactly one column, as does $N(r_{11})$.

Condition (P) holds because no three-edge path avoids a vertex of degree 3. The vertices of degree 3 are $r_{10}, r_{11}, c_{10}, c_{11}$; call the remaining 18 vertices *light*. If a three-edge path P contains a degree-3 vertex then $T(P) \geq 3 + 2 + 2 + 2 = 9$, as required. So it suffices to show that the subgraph L of G_S induced by the light vertices contains no path with three edges. The edges of L are the twelve cells of S lying in a light row and a light column, namely

$$\begin{array}{lll} (1, 1), (2, 1); & (3, 2), (4, 2); & (5, 3), (6, 3); \\ (7, 4), (7, 5); & (8, 6), (8, 7); & (9, 8), (9, 9), \end{array}$$

and these form six vertex-disjoint two-edge paths, whose centres are respectively c_1, c_2, c_3, r_7, r_8 and r_9 . Every component of L therefore has three vertices and two edges, and no three-edge path fits inside one.

By Lemma 2.2, S is a weak 4-resolving set of $K_{11} \times K_{11}$, so $\text{wdim}_4(K_{11} \times K_{11}) \leq |S| = 24$. Since (3) asserts the value $2 \cdot 11 + 1 + \lfloor 11/4 \rfloor = 25$ at $n = 11$, it is false. $\square$

4. PROOF OF THEOREM 1.2

Let $n \geq 4$ and let S be a weak 4-resolving set of $K_n \times K_n$ with $s = |S|$.

Step 1: any two row degrees sum to at least 4, and likewise for columns. Fix $i \neq i'$. The cells of S in rows i and i' occupy at most $a_i + a_{i'}$ columns, so if $a_i + a_{i'} < n$ there is a column j with $(i, j) \notin S$ and $(i', j) \notin S$. Applying Lemma 2.1 to the same-column pair $x = (i, j)$, $y = (i', j)$, which has $\alpha = 0$, gives $a_i + a_{i'} = \Delta_S(x, y) \geq 4$. Hence $a_i + a_{i'} \geq \min(n, 4) = 4$, since $n \geq 4$. The column statement is symmetric.

Step 2: we may assume every degree is at least 2. Suppose $a_{i_0} \leq 1$ for some i_0 . By Step 1 every other row has degree at least $4 - a_{i_0}$, so $s \geq a_{i_0} + (n - 1)(4 - a_{i_0})$, which is $4n - 4$ if $a_{i_0} = 0$ and $3n - 2$ if $a_{i_0} = 1$; in either case $s \geq 3n - 2$ because $n \geq 2$. Now $n \geq 4$ gives $9n \geq 33$, that is $\frac{2n}{11} + 1 \leq n - 2$, and therefore $\lceil \frac{2n}{11} \rceil < \frac{2n}{11} + 1 \leq n - 2$, so $s \geq 3n - 2 = 2n + (n - 2) \geq 2n + \lceil 2n/11 \rceil$ and we are done. The same argument applies to a column of degree at most 1. So assume from now on that $a_i \geq 2$ and $b_j \geq 2$ for all i, j , and set $E = s - 2n \geq 0$.

Step 3: heavy vertices. Call a vertex of G_S *heavy* if its degree is at least 3 and *light* if its degree is exactly 2, and let h be the number of heavy vertices. Since G_S has $2n$ vertices and s edges,

$$\sum_{v \in V(G_S)} (\deg v - 2) = 2s - 4n = 2E,$$

and every summand is ≥ 0 with the heavy ones ≥ 1 ; hence $h \leq 2E$ and

$$(7) \quad \sum_{v \text{ heavy}} \deg v = 2h + 2E.$$

Step 4: the light part is a union of short paths. Let L be the subgraph of G_S induced by the light vertices. Every vertex of L has L -degree at most 2. If L contained a path with three edges, that path would be a three-edge path of G_S with $T = 8$, contradicting Lemma 2.2(P). In particular L contains no cycle: G_S is bipartite, so a shortest cycle of L has length at least 4 and any four consecutive vertices of it form a three-edge path. Hence L is a disjoint union of paths, each with at most 3 vertices.

Step 5: counting light-heavy edges. Let k be the number of components of L . A component with $c \in \{1, 2, 3\}$ vertices has $c - 1$ edges, and the G_S -degrees of its vertices sum to $2c$, so exactly $2c - 2(c - 1) = 2$ edges leave it for heavy vertices. Summing over components, the number of light-heavy edges of G_S is exactly $2k$. Every such edge has a heavy endpoint, so by (7),

$$2k \leq \sum_{v \text{ heavy}} \deg v = 2h + 2E, \quad \text{i.e.} \quad k \leq h + E.$$

On the other hand L has $2n - h$ vertices distributed among k components of at most 3 vertices each, so $k \geq (2n - h)/3$.

Step 6. Combining, $2n - h \leq 3k \leq 3h + 3E$, whence

$$2n \leq 4h + 3E \leq 8E + 3E = 11E.$$

As E is an integer, $E \geq \lceil 2n/11 \rceil$ and $s = 2n + E \geq 2n + \lceil 2n/11 \rceil$. $\square$

Remark. The hypothesis $n \geq 4$ cannot be dropped: at $n = 3$ Step 1 yields only $a_i + a_{i'} \geq 3$, and [1] report $\text{wdim}_4(K_3 \times K_3) = 6 < 7 = 2 \cdot 3 + \lceil 6/11 \rceil$.

5. PROOF OF THEOREM 1.3

For $n = 11$: Theorem 1.1 gives $\text{wdim}_4 \leq 24$ and Theorem 1.2 gives $\text{wdim}_4 \geq 22 + \lceil 22/11 \rceil = 24$.

For $n = 12, 13, 14$ the lower bounds are $24 + \lceil 24/11 \rceil = 27$, $26 + \lceil 26/11 \rceil = 29$ and $28 + \lceil 28/11 \rceil = 31$, and the matching upper bounds are witnessed by the following sets, whose row and column degrees are all at least 2 and

which may be checked against Lemma 2.2 or directly against (1):

$$\begin{aligned} S_{12} = \{ & (1, 11), (1, 12), (2, 5), (2, 9), (3, 7), (3, 10), (3, 12), (4, 2), (4, 4), (5, 1), \\ & (5, 3), (5, 6), (6, 6), (6, 8), (7, 2), (7, 8), (7, 11), (8, 2), (8, 9), (9, 4), \\ (8) \quad & (9, 9), (10, 1), (10, 10), (11, 3), (11, 7), (12, 2), (12, 5) \}, \end{aligned}$$

$$\begin{aligned} S_{13} = \{ & (1, 6), (1, 12), (2, 7), (2, 13), (3, 1), (3, 2), (3, 5), (4, 6), (4, 10), (5, 3), \\ & (5, 5), (6, 10), (6, 12), (7, 1), (7, 7), (8, 8), (8, 11), (9, 9), (9, 10), (10, 4), \\ & (10, 8), (10, 9), (11, 3), (11, 7), (11, 11), (12, 2), (12, 4), (13, 12), \\ (9) \quad & (13, 13) \}, \end{aligned}$$

$$\begin{aligned} S_{14} = \{ & (1, 12), (1, 13), (2, 3), (2, 8), (3, 1), (3, 8), (4, 6), (4, 9), (4, 14), (5, 8), \\ & (5, 10), (6, 2), (6, 4), (7, 4), (7, 10), (8, 1), (8, 2), (8, 5), (9, 7), (9, 11), \\ & (9, 12), (10, 5), (10, 11), (11, 4), (11, 9), (12, 13), (12, 14), (13, 3), \\ (10) \quad & (13, 13), (14, 6), (14, 7) \}. \end{aligned}$$

Here $|S_{12}| = 27$, $|S_{13}| = 29$ and $|S_{14}| = 31$. This proves the first assertion, and the surmised values at these n are 28, 30, 32.

For the family, let $m \geq 1$ and let

$$S(m) = \{ (i + 11t, j + 11t) : 0 \leq t < m, (i, j) \in S \} \subseteq [11m] \times [11m]$$

be the m -fold *block-diagonal* copy of (4), of size $24m$. Its bipartite graph $G_{S(m)}$ is the disjoint union of m copies of G_S , so all its degrees are 2 or 3, it has no 4-cycle, and every three-edge path lies inside one block and hence meets a vertex of degree 3 by the argument of Section 3. Lemma 2.2 therefore gives $\text{wdim}_4(K_{11m} \times K_{11m}) \leq 24m$. Theorem 1.2 with $n = 11m$ gives the reverse inequality,

$$\text{wdim}_4(K_{11m} \times K_{11m}) \geq 22m + \left\lceil \frac{22m}{11} \right\rceil = 22m + 2m = 24m,$$

so equality holds. Finally $24m < 22m + 1 + \lfloor 11m/4 \rfloor$ for every $m \geq 1$, since $1 + \lfloor 11m/4 \rfloor - 2m \geq 1 + \frac{11m-3}{4} - 2m = \frac{3m+1}{4} > 0$. $\square$

6. SCOPE

Inequality (2) is a theorem of [1] and is not contradicted here: $24 \leq 25$, $27 \leq 28$, $29 \leq 30$ and $31 \leq 32$. What is refuted is only the surmised *equality* (3), so this note is not an erratum.

Theorem 1.2 gives $\text{wdim}_4(K_9 \times K_9) \geq 20$ and $\text{wdim}_4(K_{10} \times K_{10}) \geq 22$, against the surmised 21 and 23; combined with (2) this leaves $\text{wdim}_4(K_9 \times K_9) \in \{20, 21\}$ and $\text{wdim}_4(K_{10} \times K_{10}) \in \{22, 23\}$. We settle neither, so (3) may well be correct at the two smallest members of its own range. Theorem 1.1 nevertheless refutes (3), which is universally quantified over $n \geq 9$.

Theorem 1.3 determines $\text{wdim}_4(K_n \times K_n)$ at $n = 11, 12, 13, 14$ and at every multiple of 11. For every other $n \geq 9$ we have only $2n + \lceil 2n/11 \rceil \leq \text{wdim}_4(K_n \times K_n) \leq 2n + 1 + \lfloor n/4 \rfloor$, and we do not know which end is attained.

The lower bounds above are the hand proof of Section 4; no enumeration of candidate sets was carried out at any n , and no claim of minimality rests on a solver.

7. VERIFICATION

Every object needed above is printed in this note, and the arguments of Sections 2–5 use nothing else. The accompanying program `verify.py` reads the cell lists (4), (8)–(10) exactly as printed and uses the Python standard library only. It checks (1) against breadth-first distances in the explicitly constructed product for $3 \leq n \leq 7$; it checks Lemma 2.1 against raw distance sums over all pairs; it recomputes $\min \Delta_S$ over all pairs for each printed set and for the block-diagonal sets $S(1), S(2), S(3)$, obtaining $\min \Delta_S = 4$ in each case; and it verifies the degree sequences, the absence of a 4-cycle in G_S and the six-component structure of L . Of Theorem 1.2 it checks only the arithmetic — the chain $2n \leq 4h + 3E \leq 11E$ for $4 \leq n \leq 300$ — and not the combinatorial Steps 1, 4 and 5, which are hand proofs; likewise the block-diagonal family is machine-checked only at small m , the statement for all $m \geq 1$ being the hand proof of Section 5. The recorded run `verify.output.txt` reports 109 passing checks and states in its own output what it does not cover.

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